ASSESSMENT TOOL II- ASSIGNMENT

COURSE CODE: ITA14 **COURSE NAME: Ethical Hacking**

COURSE OUTCOME COVERED

CO1: Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

ASSIGNMENTS QUESTIONS

1. A company hires you to test its network but gives you only a one-line email authorization with no defined scope. Halfway through testing, you discover a critical vulnerability in a system that wasn't mentioned in the email. Do you exploit it to prove impact? Justify your answer using ethical hacking principles.
2. You are conducting a penetration test and discover that the client's production database contains real customer financial data. Your test plan calls for exploiting a SQL injection vulnerability. How do you proceed without causing harm, and what precautions should have been taken before the engagement began?
3. During a black-box penetration test, you accidentally trigger a denial-of-service condition on a live production server, causing an outage. Walk through what you should have done to prevent this and what your responsibility is now.
4. A client asks you to test their systems but insists you not inform their IT department, wanting to test the team's real-world detection capability. What ethical and practical risks does this create, and how would you structure the engagement to protect yourself?
5. You're asked to perform a "red team" exercise versus a "vulnerability assessment." Using a real-world scenario of a bank wanting to test physical and digital security, explain how your approach, ethics, and rules of engagement would differ between the two.

Code of Conduct:

I P. Sudharshan reddy(192372204) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Rubrics

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all data-handling requirements and operations	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate data structures and algorithms to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct code, output and complexity analysis with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. Unauthorized Vulnerability Discovery

If I discover a Critical Vulnerability outside the approved Scope, I will

1. Avoid Exploiting the Vulnerability without permission
2. Document the Vulnerability Safely without Causing damage
3. Inform the Client immediately about the discovery
4. Obtain written authorized before performing future tests.
5. follow ethical hacking such as authorized responsible disclosure and avoiding harm

Conclusion: A Vulnerability outside the approved Scope must not be exploited without proper permission.

2. Sql Injection in a production Database.

1. Avoid accessing, modifying, or deleting real financial information

2. Use a safe proof of Concept to verify the Vulnerability
3. Prefer a staging environment with dummy data for further testing.
4. Stop testing if sensitive information may be exposed.
5. Inform the Client about the Vulnerability and its risk

Precautions before testing

1. Obtain written authorization and define the testing Scope
2. Arrange backups and task accounts
3. Establish data protection rules and Confidentiality agreements
4. Prepare an incident response and recovery plan.

(3) Accidental Denial of Service outage preventive measures.

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1. Avoid aggressive testing against live production
2. Establish request limits and safe testing methods.
3. Define emergency stop conditions before starting.
4. Obtain approval for potentially disruptive tests.

Responsibilities after outage:

1. Start testing immediately.
2. Notify the client and responsible technical time.
3. Assist with system recovery and restoration of services
4. Preserve Relevant logs and document the incident honestly.

(4) Testing without informing the IT department ethical and practical risks.

1. The IT team may mistake the test for a real cyberattack.
2. Emergency response may be delayed or unnecessarily
3. Testing uncleanly may cause unexpected system.
4. Unclear authorization may create legal & risks.

measure to protect the engagement

1. Obtain written authorization form an appropriate senior authority.
2. Define the scope objectives and rules of engagement
3. Establish emergency Contact and stop procedures.

(5) Red team Exercise Vs Vulnerability Assessment

Scenario: A bank wants to test its physical and digital security.

A. Red team Exercise:

1. Simulates realistic attacks to evaluate the bank security.
2. Tests authorized physical access control and design flaws
3. Requires strict rules for attack simulations, emergency stop and protection of Customer data.

B. Vulnerability Assessment

1. Uses Approved Scanning and testing methods
2. Classifys Vulnerabilities according to their severity
3. Avoids recommendations to fix identified weakness